

## Atoms and Molecules:

- Around 500 B.C., Indian Philosopher named Maharishi Kanad postulated that matter can be divided into extremely small particles, which can't be divided further and are called as Parmanus.
- Greek Philosophers - Democritus and Leucippus gave same idea but called these indivisible particles as "Atoms".

Atom - "That can't be cut".

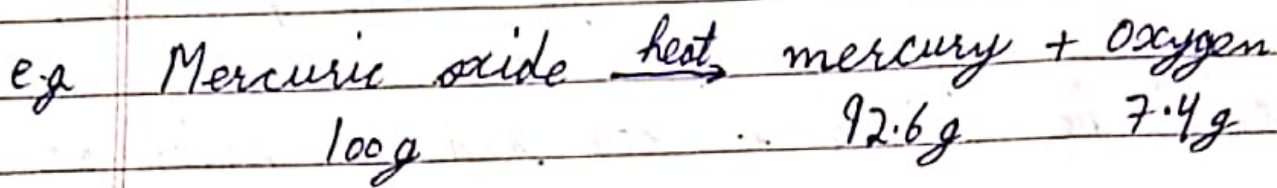
### → Laws of Chemical Combination:

- 1) Law of Conservation of mass: (By Antoine Lavoisier)
- During any physical or chemical change

the total mass of reactants is equal to total mass of products.

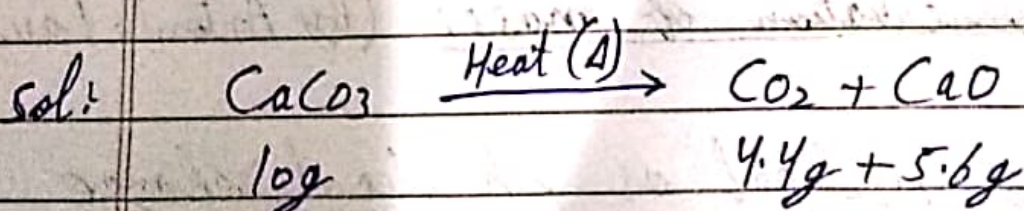
'OR'

"Mass can neither be created nor destroyed".



$$\begin{aligned} \text{Total mass of Reactants} &= \text{Total Mass of Products} \\ 100\text{g} &= 92.6 + 7.4 = 100\text{g} \end{aligned}$$

Q) If 10g of calcium carbonate ( $\text{CaCO}_3$ ) on heating gave 4.4g of carbon dioxide ( $\text{CO}_2$ ) and 5.6g of calcium oxide ( $\text{CaO}$ ). Show these observations are in agreement with law of conservation of mass?



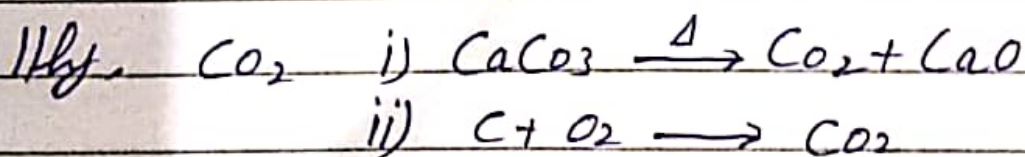
- Acc. to Law of conservation of mass of reactants = Total mass of products.

$$10g = 4.4 + 5.6$$

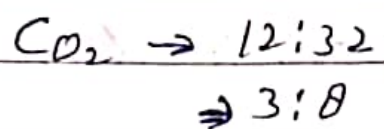
- Yes it follows Law of Conservation of mass.
- Law of Constant Composition 'OR' Definite Proportion
- \* - Proposed by Joseph Proust.
- A pure chemical compound is always made up of same elements that are combined in same proportion by mass.

e.g. Water from any source - River, Lake, Ocean, Rain, etc. is always made up of Hydrogen and Oxygen in ratio  $H_2O$

$$2:16 = 1:8$$



In case (i) and (ii)  $\text{CO}_2$  prepared by different method but elements are and ratio is same.



- Q) When 1.375g of Cupric Oxide ( $\text{CuO}$ ) was reduced on heating in a current of hydrogen the weight of copper ( $\text{Cu}$ ) that remained was 1.098g. In another experiment 1.79g of copper was dissolved in nitric acid and resulting copper nitrate converted into cupric oxide by ignition. The weight of cupric oxide formed was 1.476g. Show that these results illustrate the law of constant proportion.

sol In experiment no. 1

$$\text{Mass of CuO} = 1.375 \text{ g}$$

$$\text{Mass of Cu} = 1.098 \text{ g}$$

$$\text{Mass of oxygen} = 1.375 \text{ g} - 1.098 \text{ g} = 0.277 \text{ g}$$

$$\% \text{ of Cu} = \frac{\text{Mass of Cu}}{\text{Mass of CuO}} \times 100 = \frac{1.098 \times 100}{1.375}$$

$$= 79.85\%$$

$$= 80\%$$

$$\% \text{ of Oxygen} = \frac{\text{Mass of oxygen}}{\text{Mass of CuO}} \times 100 = \frac{0.277 \times 100}{1.375}$$

$$= 20\%$$

In experiment no. 2

$$\text{Mass of CuO} = 1.476$$

$$\text{Mass of Cu} = 1.179$$

$$\text{Mass of oxygen} = 1.476 - 1.179 = 0.297$$

$$\% \text{ of Cu} = \frac{\text{Mass of Cu}}{\text{Mass of CuO}} \times 100 = \frac{1.179 \times 100}{1.476}$$

$$= 79.8\%$$

$$= 80\%$$

$$\% \text{ of oxygen} = \frac{\text{Mass of oxygen}}{\text{Mass of CuO}} \times 100$$

$$= \frac{0.297}{1.476} \times 100$$

$$= 20\%$$

$\therefore$  Law of constant composition followed.

$\Rightarrow$  Dalton's Atomic Theory:-

- Proposed by John Dalton.

- John Dalton studied structure of matter and proposed following points:-

- 1) Matter is made up of extremely small particles called as Atoms.

- 2) Atoms of an element are similar in every respect, i.e; Mass, shape, size, etc.

- 3) Atoms of different elements are different in every respect i.e; mass, size, shape etc.
- 4) Atom is the smallest particle that takes part in a chemical reaction.
- 5) Atoms combine together in simple whole no. ratio to form molecules.
- 6) Atoms are indivisible i.e; they can't be divided further.

⇒ Limitations of Dalton's Atomic theory:

- 1) This theory did not explain why the atoms of different elements are different i.e; this theory did not talk about internal structure of Atom.

- 2) Why and how atoms combine and form molecules.
- 3) Atom is indivisible according to this theory but Atom can be divided.

⇒ Atom:- It is the smallest particle that may or may not have independent existence.

e.g. Atoms that exist independently = Na, C, K, B, etc.

Atoms that exist in combined form =  $O_2$ ,  $H_2$ ,  $N_2$ , etc.

⇒ Symbols of Atoms of different elements:-

- John Dalton proposed some nortification to represent symbols of Atoms.

Atoms symbols.

e.g. Carbon → ●

Oxygen → O

Hydrogen → ⊙

- These symbols were later discarded.

⇒ Berzelius proposed that the symbols should include letters from name of atom.

- IUPAC - International Union of pure and applied chemistry; developed following symbols:-

A) Elements that use first letter of name as symbol:-

- i) Hydrogen - H
- ii) Oxygen - O
- iii) Nitrogen - N
- iv) Carbon - C, etc.

B) First two letters in symbols:-

Aluminium - Al

Helium - He

Lithium - Li

Silicon - Si

Neon - Ne

Argon - Ar, etc.

c) First and third letter of name:-

Magnesium - Mg

Chlorine - Cl

Zinc - Zn, etc.

d) Symbols from Latin Names:-

Iron - ~~Ferr~~ Ferrum - Fe

Lead - Plumbum - Pb

Gold - Aurum - Au

Sodium - Natrium - Na

Potassium - Kalium - K

Tin - stannum - Sn

\*  $\Rightarrow$  Atomic Mass:-

- In Ancient times isolation of ~~some~~ single atoms and calculation of its mass was difficult.
- Mass of any atom was calculated with respect to some standard or reference value. So the

mass calculated is called as Relative Atomic mass.

→ Relative Atomic mass calculated with respect to some standard value.

• scale on which these masses were expressed is called as Atomic Mass scale.

→ The reference chosen for calculation of relative atomic mass.

1. Hydrogen

But mass of some atoms came out to be fractional so it was discarded.

2. Oxygen

But mass of some atoms showed variation using oxygen as standard so it was also discarded.

3. Finally IUPAC set carbon as standard and

mass of other atoms was calculated using carbon as 12 units on atomic mass scale.

⇒ Atomic Mass: It is defined as relative atomic mass of an atom with respect to C-12 taken as 12 units.

⇒ Unit: amu → Atomic mass unit

e.g. For Hydrogen = 1 amu. Oxygen = 16 amu.

⇒ Atomic Mass Unit - (amu):

• Atomic mass unit is defined as  $\frac{1}{12^{\text{th}}}$  mass of Carbon atom.

$$1 \text{ amu} = 1.66 \times 10^{-24} \text{ g}$$

⇒ Molecules:

• Molecules is defined as the smallest particle of

matter that exists freely, the properties of that substance. Individual properties of atom are not present in molecule.

e.g.  $\text{CO}_2$ ,  $\text{H}_2\text{O}$ ,  $\text{NH}_3$ , etc.

### Molecules - 2 Types

Molecule of Element

Molecule of compound

#### 1) Molecule of Element:

- Also called as ~~Homot~~ HOMOATOMIC Molecules.
- These are the molecules that are made up of same type of atoms.

e.g.  $\text{O}_2$ ,  $\text{O}_3$ ,  $\text{P}_4$  etc.

#### 2) Molecule of Compound:

- Also called as HETEROATOMIC molecules.
- These are the molecules that are made up of

different type of atoms.

e.g.  $\text{CO}_2$ ,  $\text{H}_2\text{O}$ ,  $\text{NH}_3$ , etc.

⇒ Atomicity: It is the no. of atoms present in a molecule.

- Diatomic → 2 atoms
- Triatomic → 3 atoms
- Tetraatomic → 4 atoms
- Pentaatomic → 5 atoms

| (e.g.) Homoatomic                                                                                                                              | Heteroatomic                                                                                                                    |
|------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>• Monoatomic - He, Ne, Ar, Xe, Noble gases.</li> </ul>                                                  | <ul style="list-style-type: none"> <li>• <u>Diatomic</u>: NaCl, KCl, CO</li> </ul>                                              |
| <ul style="list-style-type: none"> <li>• <u>Diatomic</u>: <math>\text{O}_2</math>, <math>\text{H}_2</math>, <math>\text{N}_2</math></li> </ul> | <ul style="list-style-type: none"> <li>• <u>Triatomic</u>: <math>\text{H}_2\text{O}</math>, <math>\text{CO}_2</math></li> </ul> |
| <ul style="list-style-type: none"> <li>• <u>Triatomic</u>: <math>\text{O}_3</math></li> </ul>                                                  | <ul style="list-style-type: none"> <li>• <u>Tetraatomic</u>: <math>\text{NH}_3</math></li> </ul>                                |
| <ul style="list-style-type: none"> <li>• <u>Tetraatomic</u>: <math>\text{P}_4</math></li> </ul>                                                |                                                                                                                                 |

⇒ Ions: Ion is defined as an atom or group of atoms carrying positive or negative charge.

e.g.  $\text{Na}^+$ ,  $\text{Cl}^-$ ,  $\text{NH}_4^+$ ,  $\text{OH}^-$ , etc.

- Ions are of two types, 1) Cation  
2) Anion.

⇒ Cation: Ion carrying positive charge.

e.g.  $\text{Na}^+$ ,  $\text{Ca}^{+2}$ ,  $\text{Al}^{+3}$ ,  $\text{NH}_4^+$ , etc.

⇒ Anion: Ion carrying negative charge.

e.g.  $\text{Cl}^-$ ,  $\text{OH}^-$ ,  $\text{O}^{-2}$ , etc.

→ Cation is formed when an atom loses electron.

→ Anion is formed when an atom gains electron.

⇒ Ions can be Monoatomic or Polyatomic.

⇒ Monoatomic Ion: Ion containing only one atom. e.g.  $\text{Na}^+$ ,  $\text{Cl}^-$ ,  $\text{O}^{-2}$ , etc.

⇒ Polyatomic Ion: Ion containing more than one atom. e.g.  $\text{CO}_3^{2-}$ ,  $\text{NH}_4^+$ ,  $\text{OH}^-$ , etc.

⇒ Molecular Mass

• Molecular mass may be defined as relative mass of a molecule with respect to mass of carbon-12 taken as 12 units.

⇒ Calculation Molecular Mass = Sum of R.A.M. Present in Molecule.

e.g.  $\text{H}_2$ .

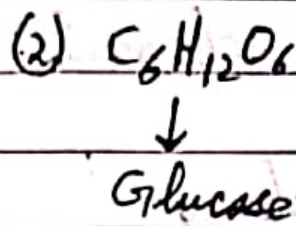
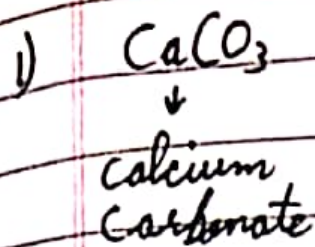
Molecular Mass =  $2 \times$  atomic mass of H.

$$= 2 \times 1$$
$$= 2 \text{ units.}$$

$\text{CO}_2$ .

$$\text{Molecular mass} = \text{C} + \text{O} + \text{O}$$
$$= 12 + 16 + 16$$
$$= 44 \text{ u.}$$

Q) Calculate molecular Mass of the following.



1) Molecular Mass of Calcium Carbonate

$$= \underset{\text{(Ca)}}{\text{Atomic mass}} + \underset{\text{(C)}}{\text{Atomic mass}} + 3 \times \text{A.M}$$

$$= 40u + 12u + 3 \times 16u$$

$$= 100u$$

$$\begin{aligned} 2) \text{C}_6\text{H}_{12}\text{O}_6 &: 6 \times \text{C} + 12 \times \text{H} + 6 \times \text{O} \\ &= 6 \times 12 + 12 \times 1 + 6 \times 16 \\ &= 180u. \end{aligned}$$

$\Rightarrow$  Valency:

- Valency is defined as combining tendency of an element.

- In case of ions Valency is defined as units.

of positive or negative charge present on ion.

⇒ Valency of some elements

| Element  | Valency |
|----------|---------|
| Hydrogen | 1       |
| Sodium   | 1       |
| Chlorine | 1       |
| Oxygen   | 2       |
| Nitrogen | 3       |
| Carbon   | 4       |

⇒ Ions Valency

|                         |   |
|-------------------------|---|
| $H^+$                   | 1 |
| $Na^+$                  | 1 |
| $Ca^{+2}$               | 2 |
| $O^{-2}$                | 2 |
| $Al^{+3}$               | 3 |
| (Hydrogen Ion) $OH^-$   | 1 |
| $Cl^-$                  | 1 |
| (Carbonate) $CO_3^{-2}$ | 2 |
| (Sulphate) $SO_4^{-2}$  | 2 |

## → Chemical Formula:

- It is defined as brief representation of a chemical compound in terms of symbols of constituents elements.

| e.g. Chemical Compound | Chemical Formula |
|------------------------|------------------|
| 1) Water               | $H_2O$           |
| 2) Carbon dioxide      | $CO_2$           |
| 3) Ammonia             | $NH_3$           |
| 4) Calcium Carbonate   | $CaCO_3$         |

## ⇒ Writing of Chemical Formula-

- Write symbol and Valency or charge of ion in manner, and apply criss cross.



- Then the valency ~~will~~ will appear in subscript

of the symbol.

→ If there is any common factor between valencies then the valencies are divisible by that common factor.

### ⇒ Important Chemical Formulas

1- Water.

- Symbol:  $\text{H} \times \text{O}$
- Valency:  $1 \times 2$
- Formula:  $\text{H}_2\text{O}$

2) Carbon dioxide

- Symbol:  $\text{C} \quad \text{O}$
- Valency:  $4 \quad 2$
- Formula:  $\text{CO}_2$

3) Methane

- Symbol:  $\text{C} \quad \text{H}$
- Valency:  $4 \quad 1$

- Formula -  $\text{CH}_4$

4) Ammonia

- Symbol -  $\text{N} \times \text{H}$

- Valency -  $3 \times 1$

- Formula -  $\text{NH}_3$

5) Carbon Monoxide

- Symbol -  $\text{C} \times \text{O}$

- Valency -  $4 \times 2$

- Formula -  $\text{CO}_2$

6) Sodium Bromide

- Symbol -  $\text{Na}^+ \times \text{Br}^-$

- Valency -  $1 \times 1$

- Formula -  $\text{NaBr}$

7) Magnesium Sulphate

- Symbol -  $\text{Mg}^{+2} \times \text{SO}_4^{-2}$

- Valency -  $2 \times 2$

- Formula -  $\text{MgSO}_4$

### 8) Sulphuric Acid

- Symbol:  $H^+ \times SO_4^{-2}$
- Valency: 1  $\times$  2
- Formula:  $H_2SO_4$

### 9) Ammonium Sulphate

- Symbol:  $NH_4^+ \times SO_4^{-2}$
- Valency: 1  $\times$  2
- Formula:  $(NH_4)_2SO_4$

⇒ Gram Atomic Mass (GAM) and Gram Molecular mass (GMM):

#### 1) Gram Atomic Mass (GAM)-

Atomic mass expressed in grams.

e.g.  $H = 1u$        $C = 12u$        $N = 14u$   
 $GAM = 1g$        $GAM = 12g$        $GAM = 14g$

#### 2) Gram Molecular Mass:- Molecular mass expressed

in grams.

e.g.  $\text{CO}_2 = 44_u$

$\text{H}_2\text{O} = 18_u$

GAMM = 44g

GMM = 18g.

⇒ Mole Concept-

- Since it was not possible to count the no. of atoms present in a sample.

- So this count was related to mass of sample.

→ It was discovered that no. of atoms present in GAM of C i.e; 12g =  $6.022 \times 10^{23}$  atoms of carbon.

→ Similarly GAM of oxygen = 16g contained:  $6.022 \times 10^{23}$  atoms of O.

→ This no.  $6.022 \times 10^{23}$  is called as Avogadro's no.

( $N_A$  or  $N$ ) after name of Amedeo Avogadro.

→ Avogadro's no. is expressed in terms of Mole.

\*⇒ 1 mole =  $6.022 \times 10^{23}$  particles.

- 1 mole of hydrogen atoms =  $6.022 \times 10^{23}$  atoms of hydrogen.

- 1 mole of water molecules =  $6.022 \times 10^{23}$  molecules of water.

- 1 mole of Sodium ion =  $6.022 \times 10^{23}$  Sodium ion.

⇒ Mole and Mass -

• Mole of ~~sub~~ atoms is defined as that amount of substance which has mass equal to Gram atomic mass or Gram molecular mass.

• 1 mole of oxygen molecule ( $O_2$ ):

→ GMM = 32g =  $6.022 \times 10^{23}$  molecules.

• 1 mole of hydrogen atoms

$$\rightarrow \text{GAM} = 1\text{g} = 6.022 \times 10^{23} \text{ atoms.}$$

⇒ Mole and no. of particles:

- One mole in terms of no. of particles is

$$1 \text{ mole} = 6.022 \times 10^{23} \text{ particles.}$$

particles  $\rightarrow$  atoms/Molecules/ions.

⇒ Mole and Volume.

- For gaseous substances:

1 mole of any gas at STP (Standard temperature and pressure) has volume equal to 22.4 litres

⇒ [STP: Temperature =  $0^\circ\text{C}$  : Pressure = 1 atm.]

⇒ Important Formulas

$$1) \text{ No. of moles} = \frac{\text{Given mass in grams}}{\text{Molecular Mass}}$$

2) No. of particles in given mass

$$\text{No. of particles} = \frac{\text{Given mass} \times \text{Avogadro's no.}}{\text{Molecular Mass}}$$

→ Numericals

Q: Calculate the no. of moles in i) 100g of iron (Fe) (Atomic mass Fe = 56u). ii) 100g of Water (Molecular mass = 18u).

Sol) (i) No. of moles =  $\frac{\text{Given mass of Fe}}{\text{Atomic mass}}$

$$\text{No. of moles} = \frac{100\text{g}}{56\text{g}} = 1.78 \text{ moles.}$$

(ii) No. of moles =  $\frac{\text{Given mass of water}}{\text{Molecular Mass}}$

$$= \frac{100\text{g}}{18} = 5.55 \text{ moles.}$$

Q: If one mole of Carbon weighs 12 gram. What is the mass of 1 atom of carbon in grams?

Sol: 1 mole =  $6.022 \times 10^{23}$  atoms = 12g.

$$6.022 \times 10^{23} \text{ atoms weigh} = 12\text{g}$$

$$\therefore 1 \text{ atom weigh} = \frac{12}{6.022 \times 10^{23}} \times 1 = 1.993 \times 10^{-23} \text{ g.}$$

Q: Calculate the mass of 0.5 moles of silver (Ag).

Atomic mass of Ag = 108u.

Sol: G.M of Ag = 108g

- 1 mole of Ag = 108g

$$\therefore 0.5 \text{ mole of Ag} = 108 \times 0.5 = 54\text{g.}$$

Q: Calculate the no. of Atoms of gold in 1g of gold. Atomic mass of Gold (Au) = 197u.

Sol: G.M. of Gold = 197 g

⇒ 197 g of Gold has =  $6.022 \times 10^{23}$  atoms.

$$1 \text{ g of Gold has} = \frac{6.022 \times 10^{23} \times 1}{197}$$

$$= 3.06 \times 10^{21} \text{ atoms.}$$

## Structure of Atom.

- Acc. to Dalton's Atomic theory atom was ~~not~~ considered to be indivisible.
- But in early 19<sup>th</sup> century some scientists challenged this theory and claimed presence of small sub atomic particles which led to discovery of these particles like electron, proton and neutron.

### ⇒ Charged Particles in Matter.

- If we rub a glass rod with silk or fur and bring it near paper ~~pieces~~ pieces, The pieces will be attracted to the rod. This indicates presence of charged particles in atoms.

## ⇒ Discovery of Electron:

Julius Plucker performed experiment using discharged tube. When the pressure of air in the discharged tube is reduced to  $0.001 \text{ mm}$  of Hg and a high voltage is applied to the electrodes. Fluorescence was observed in the walls of discharged tubes at the end opposite to cathode. It was observed that some invisible rays were formed at the cathode which on striking the glass tube emitted light.

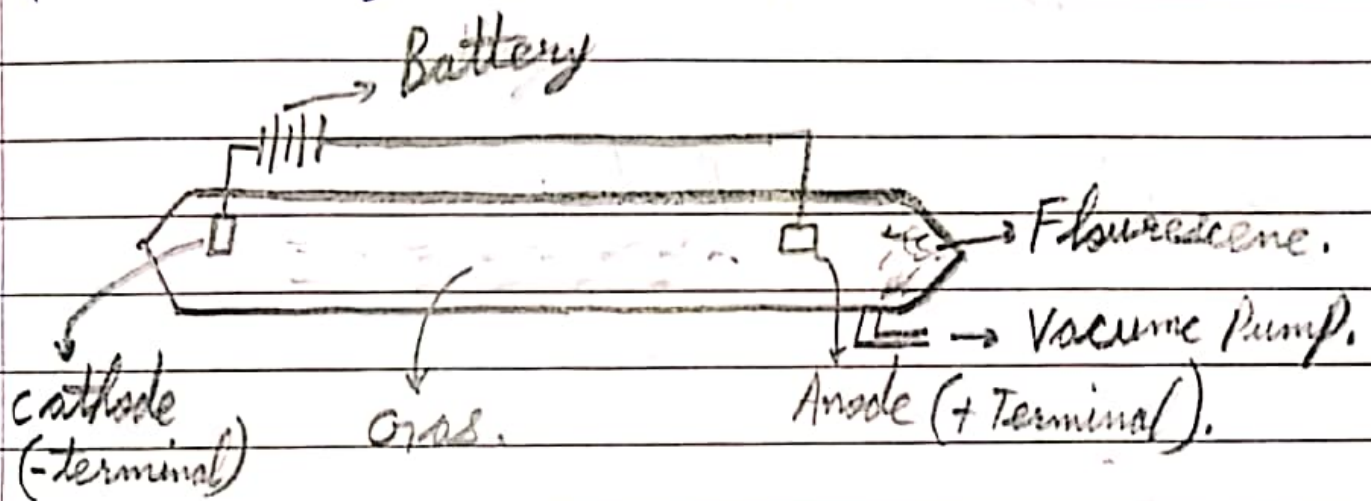
Since these rays are formed at cathode so they are called as cathode rays.

When an electric field is applied in the

path of cathode rays they are deflected towards the positive plate of electric field.

- This shows the cathode rays are made up of negatively charged particles called as Electrons.

(Discovered by J.J. Thomson).



Discharged tube Electron.

~~Mass of electron =  $9 \times 10^{-31}$  kg~~

- Mass of electron: Mass of electron is  $\frac{1}{1840}$  that of mass of hydrogen atom.

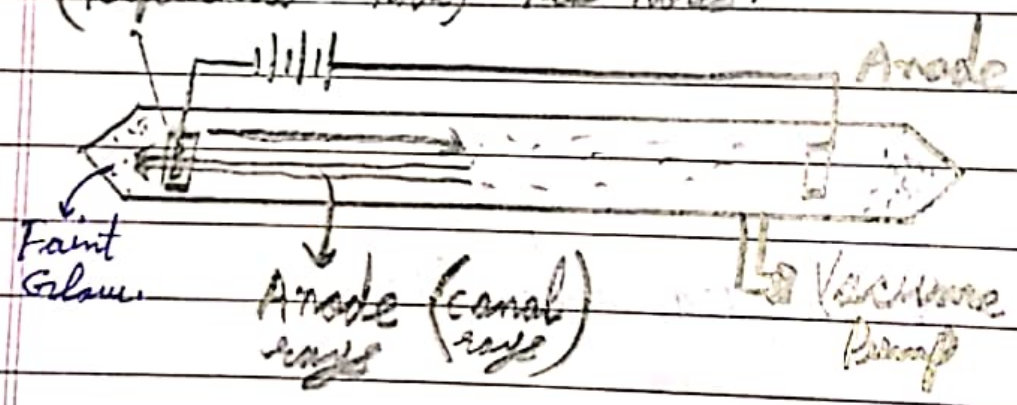
$\Rightarrow$  Absolute mass of electron is  $9 \times 10^{-31}$  kg,  $9 \times 10^{-28}$  g.

### ⇒ Charge of Electron:-

∴ Electron carries charge of  $1.6 \times 10^{-19}$  coulombs of negative charge.

→ This is the smallest value of negative charge as also c/a Unit Negative charge.

### ⇒ Discovery of Proton:- (perforated cathode) → has holes.



⇒ Eugen Goldstein experimentally proved the existence of positive charge in matter.

He used perforated cathode in discharged

tube along with air at very low pressure about  $0.001$  mm of Hg. When a high voltage of about  $10000$  Volts was applied to this cathode in the discharged tube, a faint glow was observed behind the perforated cathode.

The rays were formed in between the Cathode and Anode and these rays that produced the Faint Glow were called as Anode rays or positive or canal rays.

- When electric field is applied in the path of Anode rays, they are deflected towards the negative plate. Which shows that these rays are positively charged and are made

of up of positively charged particles and are c/a Protons.

⇒ Mass of Proton:-

- Mass of proton is ~~also~~ almost equal to mass of hydrogen atom.

★ → Absolute mass of proton =  $1.6 \times 10^{-24} \text{ g}$   
 $= 1.6 \times 10^{-27} \text{ kg}$ .

⇒ Charge of Proton:-

- Charge of proton =  $1.6 \times 10^{-19}$  coulombs of positive charge.

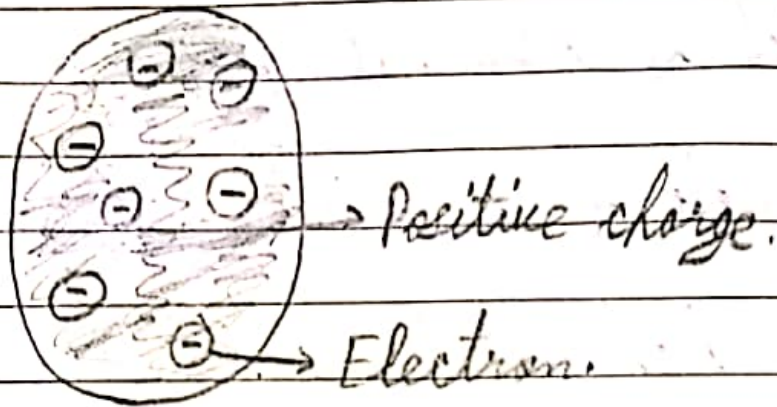
- Which is considered as Unit positive charge.

⇒ Models of Atom:-

⇒ 1) Watermelon or Raisin Pudding Model.

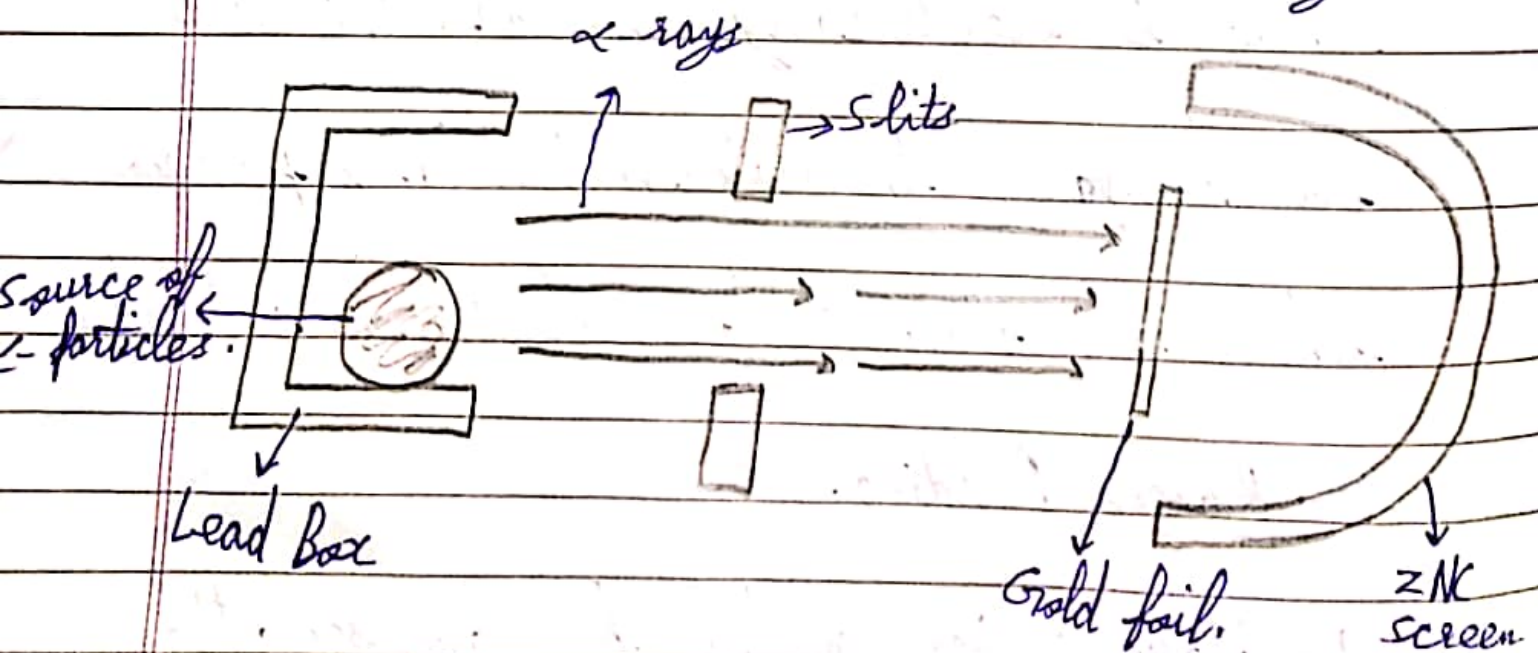
- Proposed by J.J. Thomson.
- J.J. Thomson was first to propose a detailed model of Atom. He proposed that an Atom consists of a uniform sphere of positive charge in which electrons are embedded i.e; distributed in ~~the~~ the sphere.
- The negative and positive charges are uniform in magnitude that shows the atom as a whole is electrically neutral. This model was compared to watermelon (Pink positive sphere and black seeds as electrons). and also to Raisin Pudding.
- ⇒ Limitation:- This model had no scientific

proof of arrangement of electron and proton, so it was rejected.



## 2) Rutherford's Model of Atom:

Rutherford proposed this model after performing an experiment called as  $\alpha$ -scattering experiment.



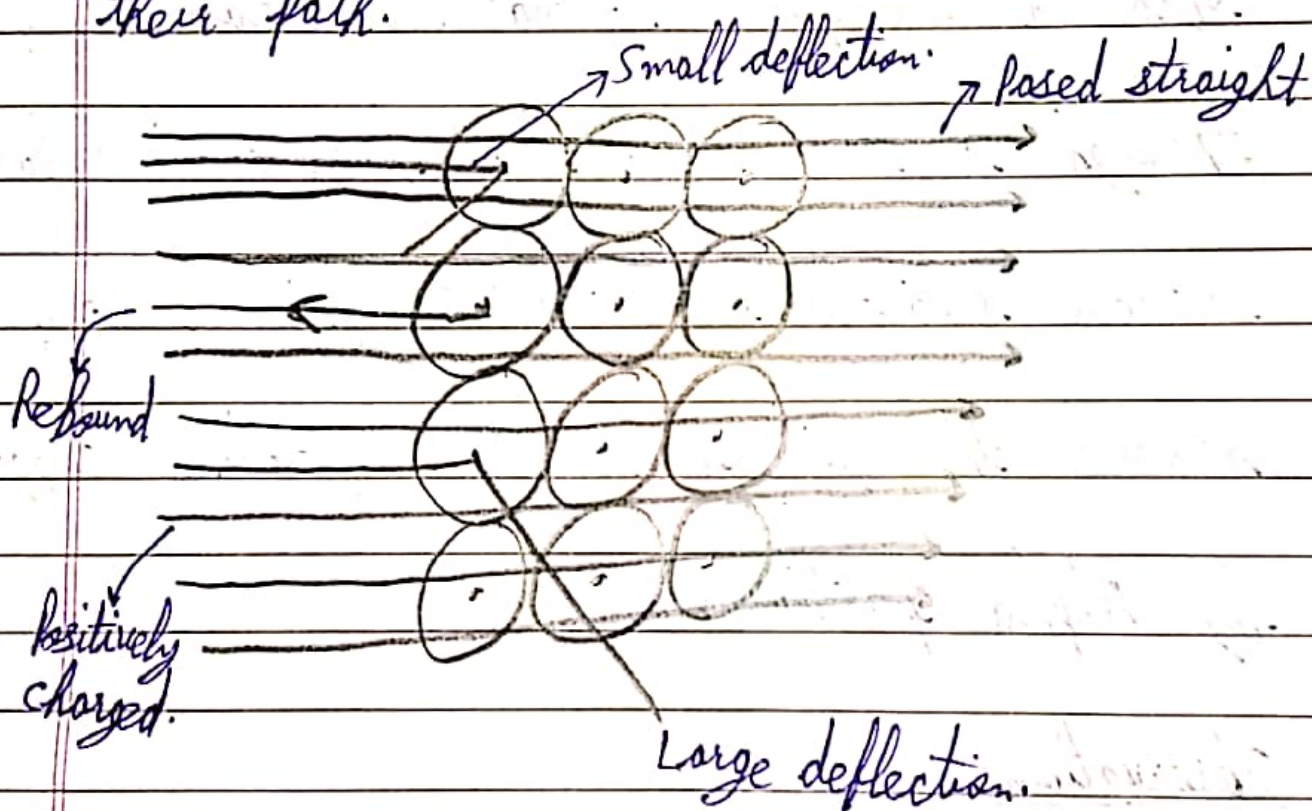
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⇒ Experiment: A thin sheet of Gold ( $4 \times 10^{-5} \text{ cm}$ ) was bombarded with Alpha particles (positively charged particles). The Alpha particles were obtained in form of narrow beam by passing through a slit in a lead box. A circular screen coated with ZNS-sulphide was placed around the foil to detect deflection suffered by Alpha particles.

⇒ Observations:

- 1) Most of the Alpha Particles (99%) passed straight through the gold foil without any deflection from their original path.

- 2) A few Alpha particles were deflected through small and large angles.
- 3) A few Alpha particles rebounded completely on hitting the gold foil and turned back their path.



### → Explanation:

- 1) Since most of the Alpha particles pass straight through the gold foil without any deflection,

It shows there is a lot of empty space in an atom.

- 2) Some of the Alpha particles are deflected through small and large angles, which shows that there is a centre of positive charge in an atom, which repel the positively charged particles.
- 3) Very few Alpha particles rebound on hitting the gold foil, ~~which~~ which shows that the ~~nucleus~~ <sup>nucleus</sup> is very dense and hard & which does not allow Alpha particles to pass through it.

⇒ Rutherford's Model of Atom.

- 1) An Atom consists of two parts; Nucleus and extra part.

